Clase 1

Sobre la evaluación:

Recuerden que hay Mini Proyectos, Mini labs & Proyectos Finales, el proyecto final es para ca. el último cuarto del semestre (más o menos el último mes). Los minis son largos & usualmente se dan 15 días para hacerlo. Los Mini Labs son opcionales, pero bien útiles si los quieren para perderle el miedo a la programación.

Ahora, yo califico todo, sé que se copian. La verdad es que los problemas los desarrollo yo entonces sé que es muy probable que para la gran mayoría no estén las soluciones en internet. Ahora bien, sé que pueden & van a copiar, no me molesta, pero lo que sí me molesta es no ver la cita al trabajo de la otra persona, el plagio sí es un problema para mí. Entonces, si copian, escríbanlo, no voy a penalizarlos pero sí si no dan los créditos correspondientes

NOTA

Lo que sí no voy a aceptar es que me pongan una foto del trabajo de la otra persona. Creo que al menos, usando sus pro manos para escribirlo / typearlo hace que aprendan algo.

• Sobre el temario

Estos cursos son complicados pues requieren muchos precedentes en particular EDOs y Álgebra lineal, también cálculo, especialmente dibujar curvas y un poco/mucho de pensamiento geométrico.
Si han llevado materias de otras carreras ó estudiado algo de transdisciplina/multidisciplina más o menos ya se van a saber este juego. Si quieren aprender a como jugar un poco con las herramientas que nos brindan las matemáticas y la realidad les van a divertir.

Es una materia que, va a parecer trivial, pero que si la dejan & no le presten atención les puede dar una sorpresa por las sutilezas. No va a parecer un curso de matemáticas en el sentido estandar pues no habrá mucho de demostración-teorema-corolario sino usaremos ciertos resultados para poder avanzar. Hay que usar mucha intuición, pensamiento geométrico y dibujos.
Lo primero que vamos a trabajar y repasar es sistemas dinámicos no-lineales. Para esto vamos con un poco de historia.

Historical Overview.

- 1666 Newton – Cálculo, Órbitos (Elipses (Kepler)) – Planetas,
Leyes de la óptica,
Problema de 2-cuerpos.
→ ¿Qué pasa con 3-Planetas, N-Planetas

"Ningún Problema me ha hecho doler tanto la cabeza como el problema de el Sol – Tierra – Luna"
problema de la luna

Carta a Halley.

200 AÑOS Euler, Gauß fallan.

- Finals de 1800: Poincaré, demuestra/da argumentos de la dificultad de poder encontrar una forma cerrada.

→ Se introduce la idea del plano fase.

→ Sus ideas son lo que ahora llamamos <u>caos</u>

<u>CAOS</u>: Comportamiento aperiódico que en un sistema determinista que exhibe "sensibilidad a condiciones iniciales"

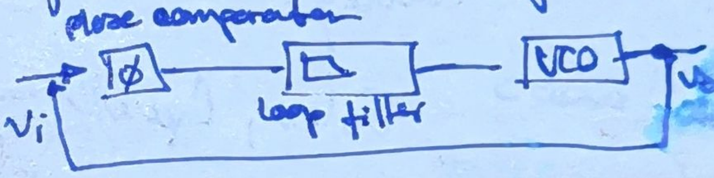
Fact: Poincaré era terrible dibujando, tan terrible que en los exámenes de ingreso en la prueba de Dibujo mecánico sacó un 0. Sin embargo su aproximación a los sistemas era muy visual, muy geométrica. A pesar de que en sus artículos no dibujaba nada, pero describía cosas. Al no dibujar sus trabajos pasaron desapercibidos.

→ Además de todo, sus trabajos pasaron un poco desapercibidos pues la acción no estaba en la mecánica clásica en la época sino en la mecánica cuántica, la relatividad (1939 - 1941 - WWII)

1920s - 1950s : Non linear oscilators (Diodes) — Physics & Engineering

↳ Radio - Vacuum tube - Non liner oscillator

- radars - phase locked loops
- Lasers.




1950 : Computer invented. (War motivated).

1954 : Artículo de ~~las~~ Lonchister.

1957 : Sputnik. 1

1960's : ~~Lorenz~~ Lorenz @ MIT (Mat & Meteorologyst).

↳ Chotic Systems on a model of convection in the athmosphere

He was interested on wether forecasting, while simplifiying the system. fond this behavior

1963 "Deterministic non-periodic flow".

↳ Desde este artículo se debió iniciar la revolución del caos pero fue ignorado

Partly because it was published on a metheorology "The journal of atmospheric sciences" (no matematicians read this) & the dynamics physicist thought it was a meteorology model.

- there is a [illegible] very smaller [illegible]

KAM (Kolmogorov - Arnold - Moser)

↳ Very mathematical course.

- 1975 - Bob May. (Pop. biologis). Chaos in iterated maps

$$x_{n+1} = f(x_n)$$

↳ 1976 "Simple Mathematical Models with very complicaded Dynamics" was published on Nature.

↳ NEW MATHEMATICS

- Evangelical Plea: Stop teaching only linear mathematics. "Si permitimos a los sistemas ser no-lineals es donde aparece la magia".

- Logistic Eqn in population biology.

- Mandelbrot: Fractals - Very obseed with the formas on nature.

- Winfree: Non-linear oscillators in biology → topology.

- Ruelle & Takens: Llevaron el caos al problema de la turbulencia & Navier-Stokes.

¿Es la turb. ~~el caos~~ una forma de caos?

¿Son lo mismo?

~1978 Feigenbaum (Physicist).

- Una conexión entre el mapeo logístico, mapeos iterados & trabajos en fenómenos de fase.

- "Universal route to chaos" - How to progress to chaos.

- Renormalization group & phase transitions in statistical physics.

~1980s - Chaos, non-linear dynamics, fractals HOT!

1989 - Chaos: Making a new science - James Gleick
a way to take the mathematics to the general public | Best sellers
↓
Tazas & playeras
↓
1993 - Jurassic Park | "La mariposa"
... Es difícil de explicar que va a pasar con los dinos.

- Experimental confirmation of chaos theory.

~1990s - Engineering Aplication of Chaos
- Encoding signals,
↳ Complex Systems.

Statistical Physics
+
Classical Mechanics

• [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]
[illegible] [illegible] números largos, incluso se vio el
[illegible] conocimiento con la [illegible] ... Cuerpos [illegible]
[illegible] mover a un satélite que orbitaba el sol para
[illegible] caer en un cometa ..., ISEE - 3
(Internationa Sun-Earth Explorer - 3)

1985 Giacobini - Ziner en [illegible]

ICE
International
Comet
Explorer.

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Logical Structure of dynamics. (De lo que cambia respecto a algo).

D.E. $\bar{x}' = \bar{F}(\bar{x})$ $\bar{x} \in \mathbb{R}^n$, $F: \mathbb{R}^n \to \mathbb{R}^n$ $\mathbb{R}^n$: Phase Space.

$$x_1' = f_1(x_1, x_2, x_3, \dots, x_n)$$
$$\vdots$$
$$x_n' = f_n(x_1, x_2, x_3, \dots, x_n).$$

Decimos que un sistema es lineal si todos los $x_i$ en el lado derecho son productos de primer orden.

$x_i^2$ $x_i x_j$ $\sin(x_i)$ $\to$ No-lineal.

Simple Examples

$m\ddot{x} + kx$

oscilador armónico

Floquet theory

sistema lineal

segundo orden.

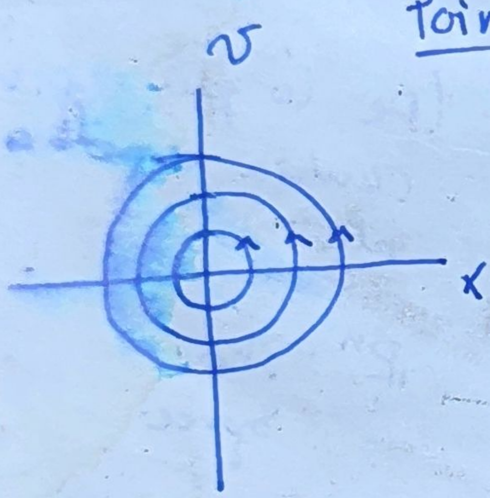
$\ddot{x} + \sin(x) = 0 \quad \Rightarrow \quad \dot{x} = v$

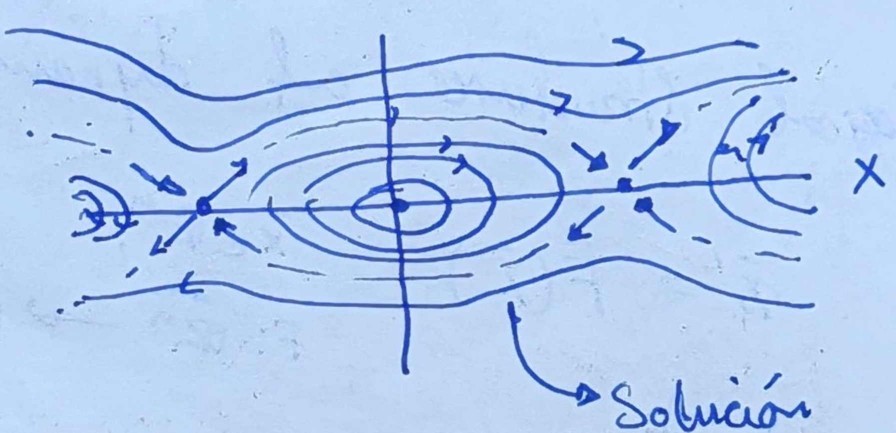
$\dot{v} = -\sin(x)$

péndulo

↳ Sistema no-lineal de segundo orden.

Poincaré idea




Solucion $(x(t), v(t)) \longrightarrow$ trayectoria en el punto moviendose en una espacio $-xv$ (Kinda - Es una parametrización de la curva).

"Cómo se verán las soluciones sin resolver el problema, sin encontrar una solución explícita".

Retrato Fase ≡ Una imágen de todas las posibles trayectorias diferentes, se encuentran sin resolver analiticamente los sistemas.

Hands-On Example.

Pensemos en 1D systems $\dot{x} = f(x)$

La idea geométrica es un flujo en una línea.

Pensemos en el sistema:

$\dot{x} = \cancel{\sin(x)}\ ax$ , si queremos resolverla analíticamente

$x_0 = x(t=0)$ entonces proponemos

$$\frac{dx}{dt} = ax \underset{1/x}{\Rightarrow} \frac{1}{x}\frac{dx}{dt} = a$$

$$\underset{\int dt}{\Rightarrow} \int \frac{1}{x}\frac{dx}{dt}\, dt = \int a\, dt$$

$$\Rightarrow \hat{x} = x \quad \text{ó} \quad d\hat{x} = \frac{dx}{dt}\, dt \qquad \int \frac{1}{x}\frac{dx}{dt}\, dt = \int \frac{1}{\hat{x}}\, d\hat{x} = \ln(\hat{x})$$

$$\Rightarrow \ln(x) = \int \frac{1}{x}\frac{dx}{dt}\, dt = \int a\, dt = a + cte.$$

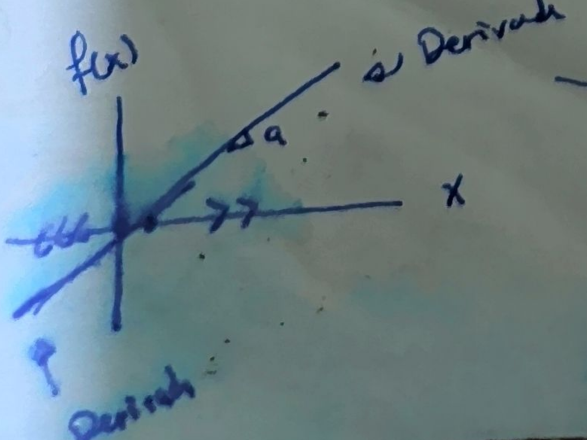
$$\Rightarrow \ln(x) = at + cte \Rightarrow x(t) = e^{at + cte} = k e^{at}$$

→ solución general

$$\Rightarrow x(0) = x_0 \to k e^{a0} = x_0 \Rightarrow k = x_0$$

$$\Rightarrow x(t) = x_0 e^{at} \text{ → solución particular.}$$

¿Qué nos dice esto del sistema?




→ la derivada siempre crece
decrece,

Hagamos un ejemplo más [illegible]

$x' = \sin(x)$ , $x_0 = x(t=0)$.

$$\frac{dx}{dt} = \sin(x) \overset{1/\sin(x)}{\Rightarrow} \int \frac{1}{\sin(x)}\frac{dx}{dt}\,dt = \int dt = t + cte$$

$$\Rightarrow \int \frac{1}{\sin(\hat{x})}\,d\hat{x} = \int dt \quad ; \quad \frac{1}{\sin(x)} = \csc(x)$$

$$\Rightarrow \int \csc(\hat{x})\,d\hat{x} = -\ln|\csc(x) + \cot(x)|$$

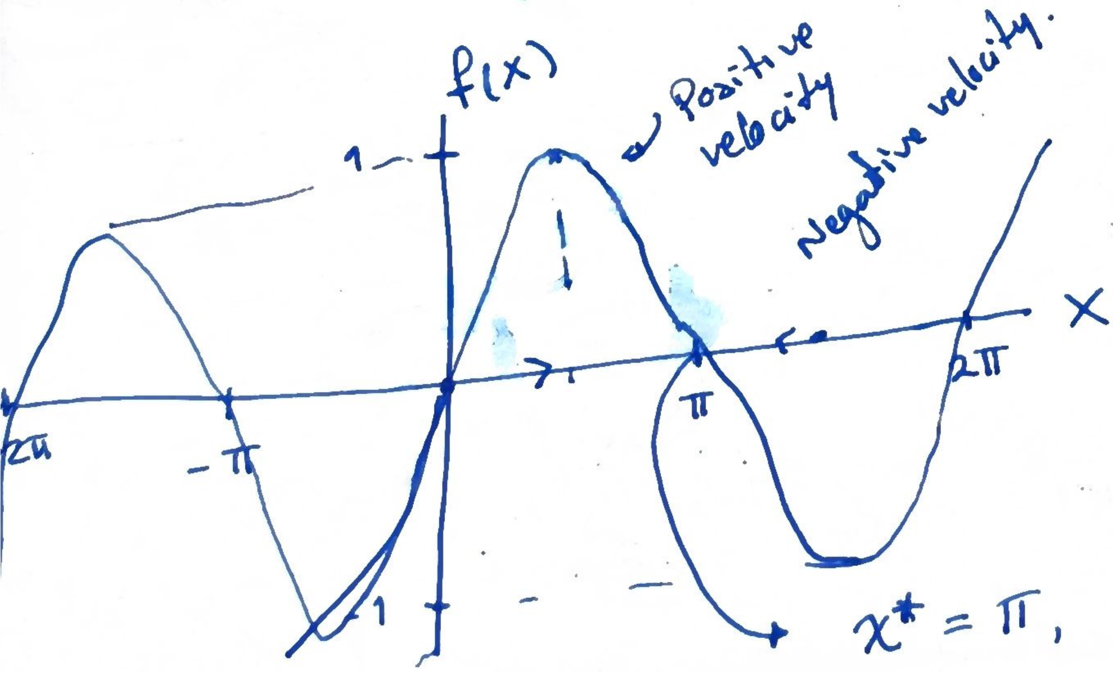
$$\Rightarrow -\ln|\csc(x) + \cot(x)| = t + cte \quad \rightarrow \text{Solución general.}$$

$t = 0 \rightarrow x = x_0$

$$\Rightarrow t = \ln\left|\frac{\csc x_0 + \cot x_0}{\csc x + \cot x}\right|. \quad (?)$$

¿Qué pasa si $x_0 = \pi/4$?

¿Cuál es el comportamiento $t \rightarrow \infty$?




¿Qué pasa con $x_0 = \pi$?

$x^* = \pi$, $x^* = n\pi$.

cero velocity.

if $x_0 = \frac{\pi}{4}$ ent $x \rightarrow \pi$, $t \rightarrow \infty$


$x - t$

# One dimensional Flows

We know that the systems are of the form $\vec{x}' = \vec{F}(\bar{x})$
$$\begin{cases} \dot{x}_1 = f_1(x_1, x_2, x_3, \dots, x_n) \\ \vdots \\ \dot{x}_n = f_n(x_1, x_2, \dots, x_n). \end{cases}$$

If we limit ourselves to 1D then we are on the terrain of first order system / 1D sys.

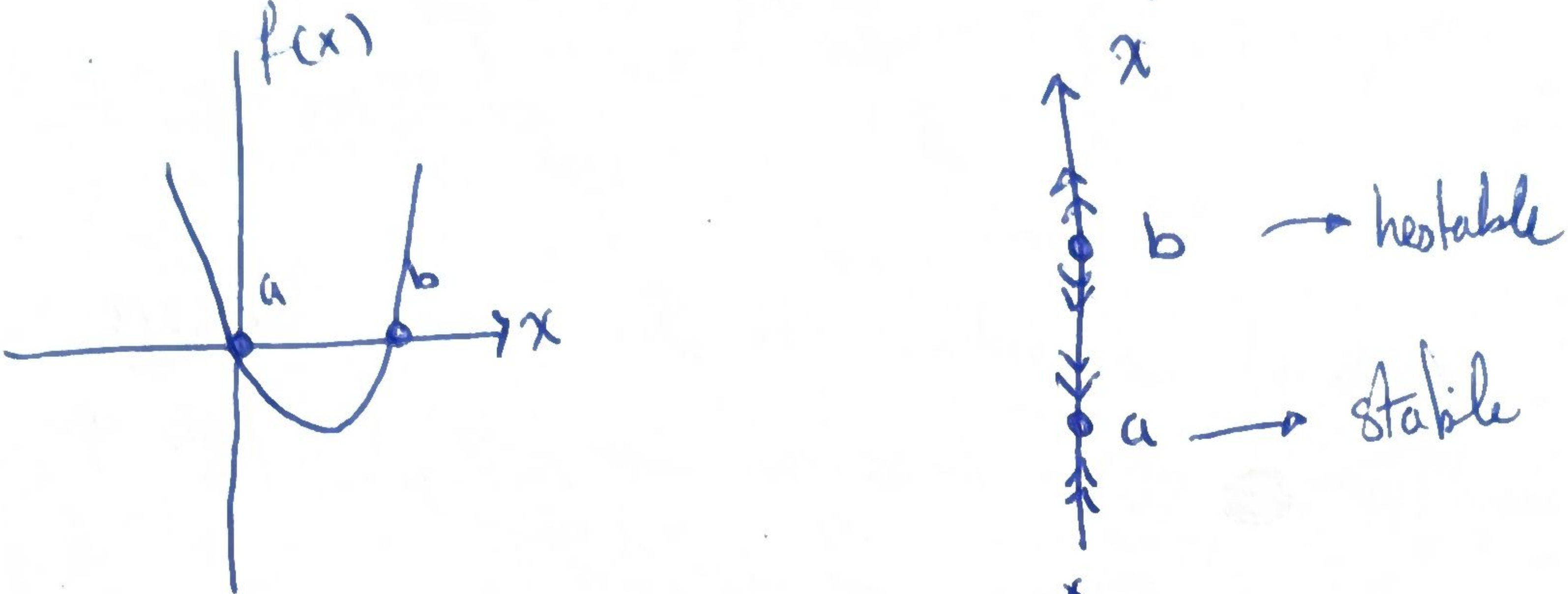
NOTE:

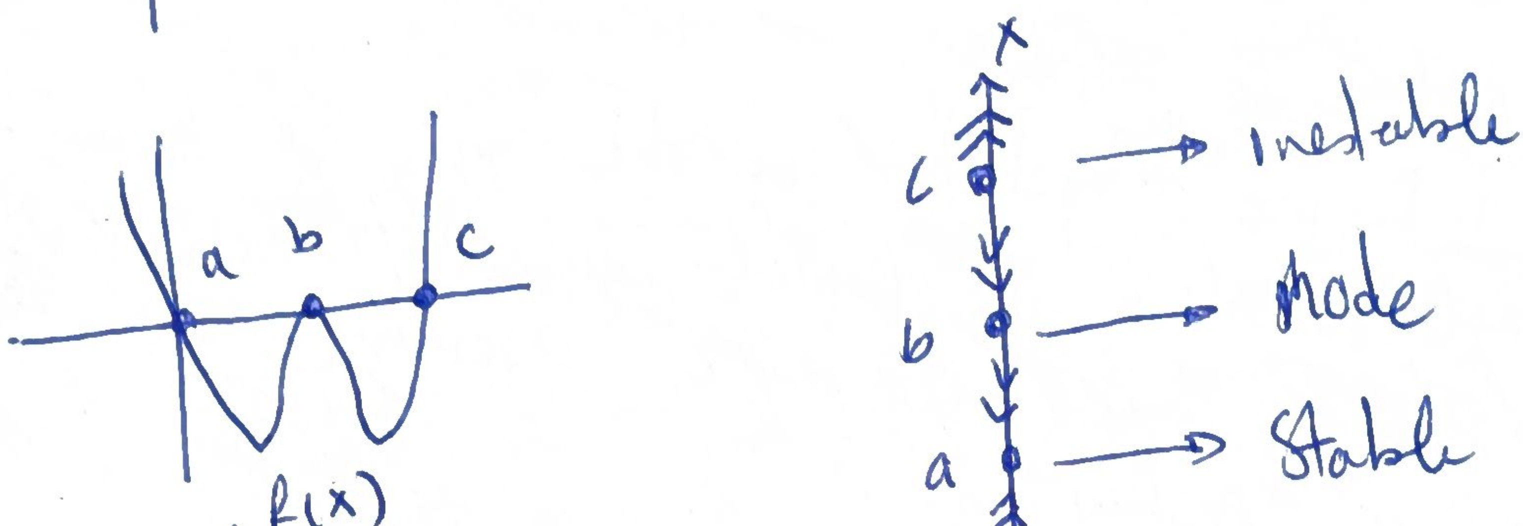
1. I use system in the sense of dynamical system, not in the classical sense of Equations systems, thus a single equation could be a "system"

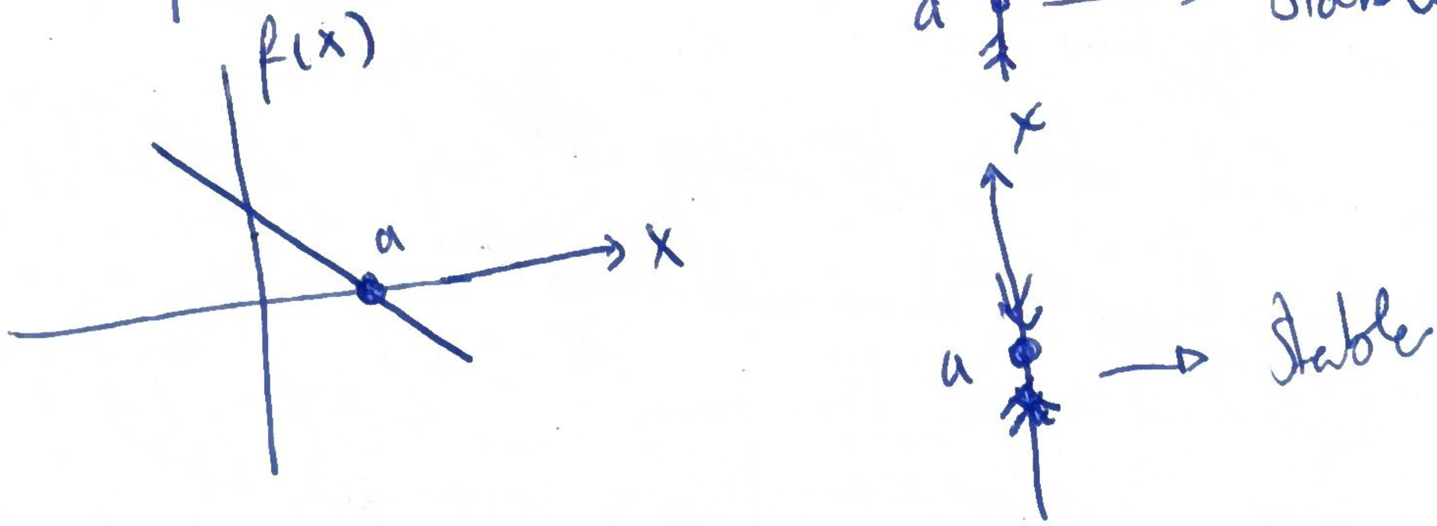
2. We won't allow to depend $f$ explictly on the time. "Non-autonomous" equations or time-dependant of the form $x' = f(x,t)$ are more complicated because they need two pieces of information "$x$ & $t$" thus $\dot{x} = f(x,t)$ should be regarded as two dimensional system or second order.

# Fixed Points & stability.

The ideas of the last section/class could be extended to any one-dimensional system $\dot{x} = f(x)$



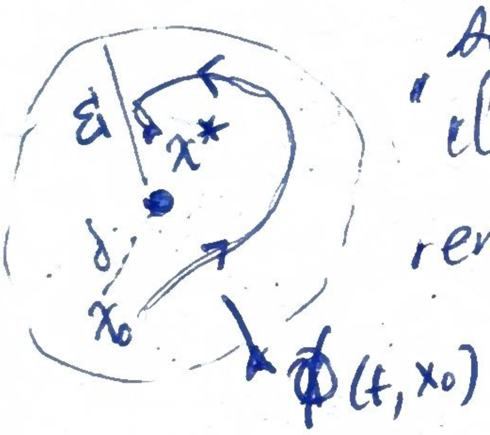




Let's notice that if the system $x' = f(x)$ ~~∃x'~~ we say that the equilibrium point is the defined as ~~*~~ any $x^*$ st. $\boxed{f(x^*) = 0}$

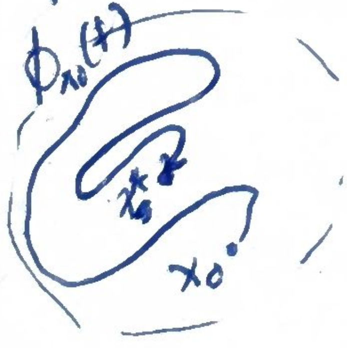
then also we could define stability (in the sense of Lyapunov) as

① $x^*$ is said to be stable if for every $\varepsilon > 0$ $\exists$ $\delta > 0$ such that if $\| x(0) - x^* \| < \delta$ then $\forall t \geq 0$ we have $\| x(t) - x^* \| < \varepsilon$




so basically if the solution starts "close enough" to the equilibrium remains "close *enough" forever

② $x^*$ is said to be asymptotically stabl if it's Lyapunov Stable & $\exists$ $\delta > 0$ such that if $\| x(0) - x^* \| < \delta$ then $\lim_{t \to \infty} \| x(t) - x^* \| = 0$

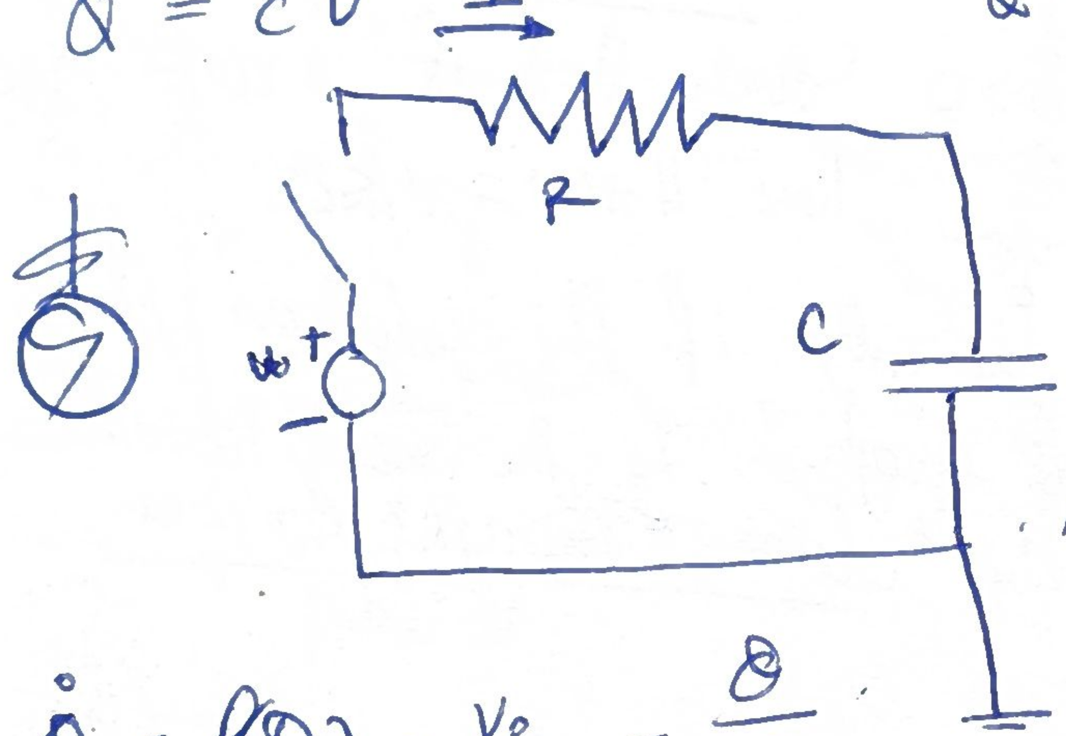



Basically means that solutions that starts close enough not only remains close enough forever but also converges to the equilibrium //

↳ What about the time? Converges linearly, polinomi, exp, etc.

ider the electrical circuit. A resistor $R$ and a capacitor $C$ are in series with batery of constant voltage $V_0$ (DC)
Suppose the switch is close a $t=0$, and there is no charge capacitor initally. Let $Q(t)$ denote the charge on the capacitor at time $t \geq 0$

$Q = CV$ $\qquad\qquad$ $\dot{Q} = I$



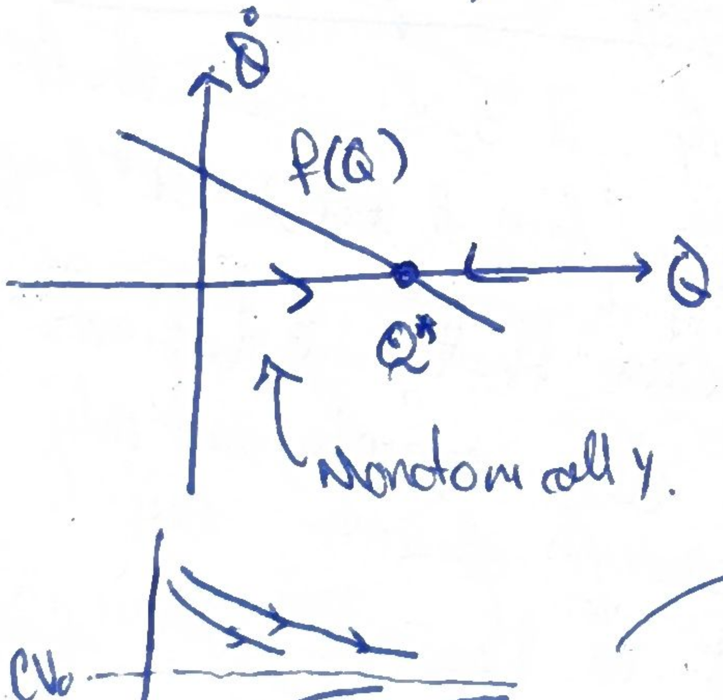

$$V_0 = \frac{Q}{C} + RI$$
$$= \frac{Q}{C} + R\dot{Q}$$
$$R\dot{Q} + Q/C - V_0 = 0$$

$$\dot{Q} = f(Q) = \frac{V_0}{R} - \frac{Q}{RC}$$

¿ $\dot{Q} = 0$ ?

$Q^*$ ?

$Q^* = CV_0$

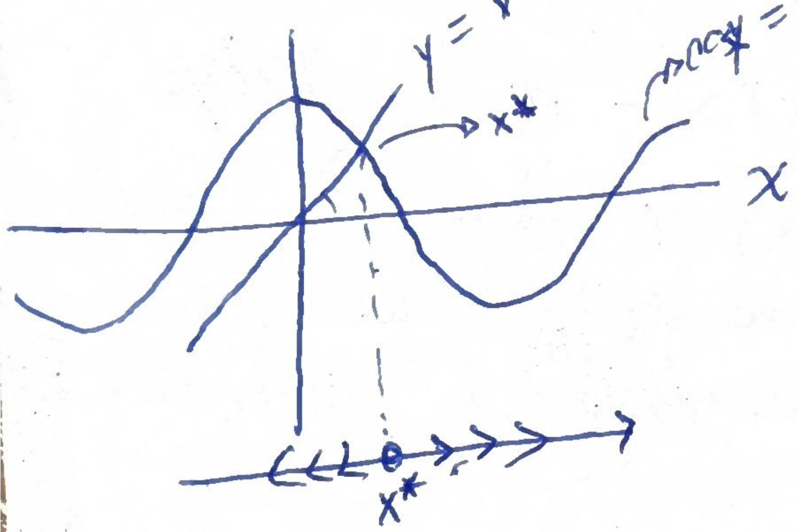




→ $\dot{Q}$ decreases linearly as $Q(t)$ approches the fixed point.

EXAMPLE:
Sketch the phase portrait of $x' = x - \cos x$ and determine the stability of all the fixed points.

<u>SOL</u> We could draw the graph $x - \cos x$ but we depend into looking what $x - \cos x$ looks like. There is an easier sketching, which exploits the fact that we know how to graph $y = x$ & $y = \cos x$.




Let's notice that

$x^* - \cos x^* = 0$

$\Rightarrow x^* = \cos x^*$

$\hookrightarrow f(x^*) = 0$

Now if $x > \cos(x)$

then $x - \cos(x) > 0$

$x < \cos(x) \Rightarrow x - \cos(x) < 0$

we have the stability without finding the explicit equation.

Popn Growth.

the simplest model we could talk of is $N' = rN$, where $N(t)$ is the population at time $t$, an $r$ is the growth rate. $(r>0)$.

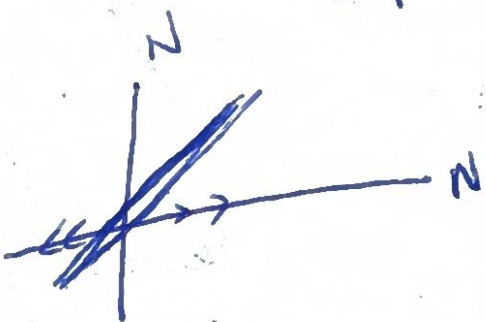
$$N' = \frac{dN}{dt} = rN \Rightarrow \int \frac{1}{N}\frac{dN}{dt}dt = \int r\,dt$$

$$\Rightarrow \int \frac{1}{N}\frac{dN}{dt}dt = \int \frac{1}{\hat{N}}d\hat{N} = \ln(\hat{N}) = rt + cte.$$

$\hat{N} = N$

$\frac{dN}{dt}dt = d\hat{N}$

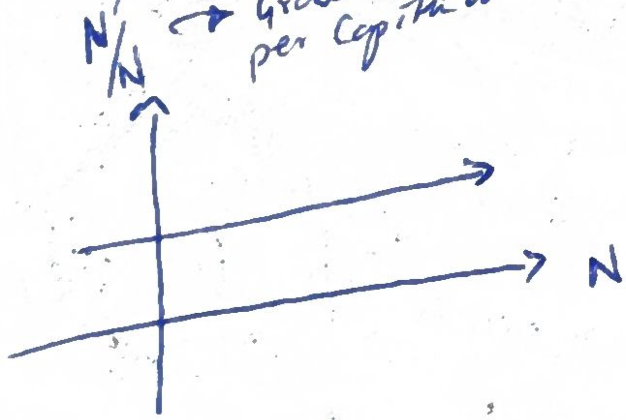
$$\rightarrow \hat{N}(t) = N(t) = e^{(cte + rt)}$$
$$= e^{cte}e^{rt}$$
$$= N_0 e^{rt}$$

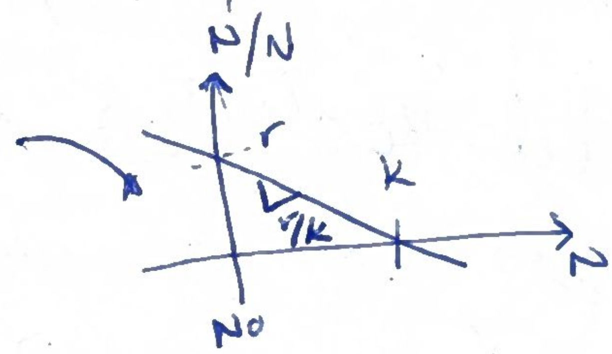



Notice that $N(t=0) = N_0$

$$\boxed{\frac{N'}{N} = r}$$

$N'/N \rightarrow$ Growth rate per capita wrt t.



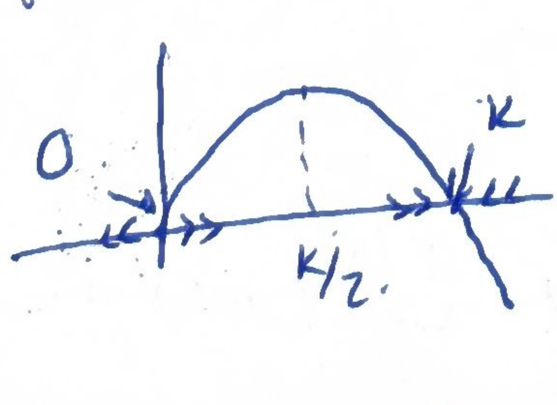




we know that the popn. starts to fight for resorses when there is over crow population. let's understand resources as smthg big.

$\frac{N'}{N} = \cancel{rN}$

$$\frac{N'}{N} = r - \frac{rN}{K}$$

$$N' = rN\left(1 - \frac{N}{K}\right)$$

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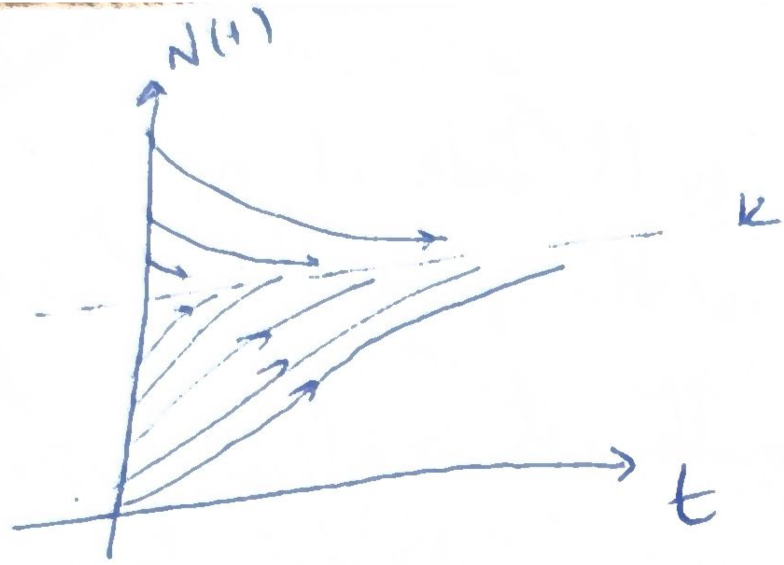



This is a metaphor!

Krebs (1972, pg. 190-200)

↳ bacteria, Yeast.

↳ (go) Petri

Krebs (1972)

Fruit flies*, escarabajos.

eggs > Larva > pupae > adults

Fluctuations ← due to age-structure and time delayed effects of overcrowding pop.n → Many eggs don't affect one a short time.

## Linear Stability Analysis

Let ~~the~~ $x' = f(x)$ be our system and ~~$\eta(t)$~~ $- x^*$ a point such that $f(x^*) = 0$

an ~~let~~ let $\eta(t) = x(t) - x^*$ a small perturbation away from $x^*$. Let's measure if the perturbation get's closer or goes away.

$$\eta'(t) = (x(t) - x^*)' = x' \Rightarrow \eta' = x' = f(x)$$
$$= f(x^* + \eta)$$

Taylor expansion

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$f(x^*+\eta) = f(x^*) + f'(x^*)(\overbrace{x - x^*}^{\eta}) + \underbrace{O(\eta^2)}_{\to 0}$$

$\|x^* - x(t)\| < \varepsilon \ll 1$

$$\to \underbrace{f(x^*)}_{\to 0} + \eta f'(x^*) + O(\eta^2)$$

$$f(x^* + \eta) = \eta f'(x^*) + O(\eta^2)$$

Now, if $f'(x^*) \neq 0$ and $O(\eta^2)$ son despreciables

$\eta' \approx \eta f'(x^*) \to$ linearization about $x^*$

this means that a perturbation grows exponentially if $f'(x) > 0$ and decays if $f'(x) < 0$. If $f'(x) = 0$ then we cannot ignore $O(\eta^2)$ terms and we need to perform non-linear analysis to determine stability.

NOTICE THAT NOW WE CAN KNOW HOW stable/unstable a point $x^*$ is. this is quantified on the value $f'(x)$ (Not only the sign is important). Also notice that $1/|f'(x)|$ is called "characteristic time scale" and it's basically the ~~required~~ time required for $x(t)$ to vary significantly around $x^*$.

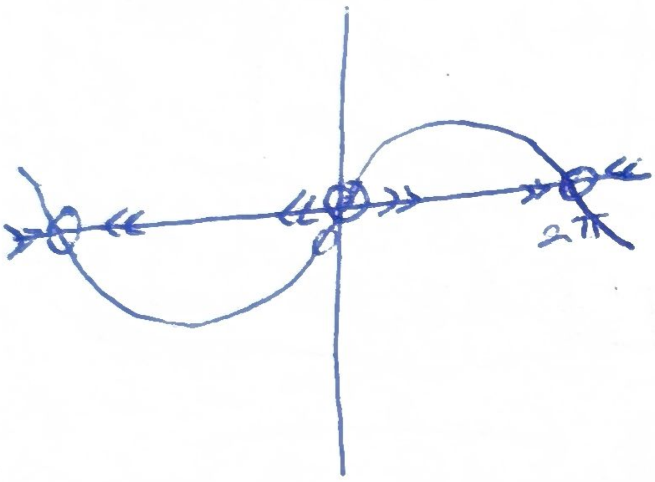
Example

Determine the stability & fixed points of $x' = \sin x$.

sol

$$x' = f(x) = \sin x = 0$$

$$x^*_k = k\pi, \quad k \in \mathbb{Z}$$



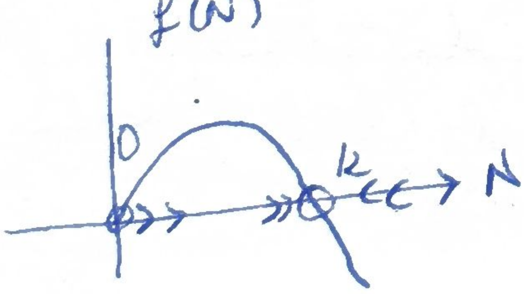

$$f'(x) = \cos(x)$$

$$f'(x^*) = \begin{cases} 1 & \text{if } k \text{ is even} \to \text{unstable} \\ -1 & \text{if } k \text{ is odd} \to \text{Stable.} \end{cases}$$

---

Ex. Find the eq. points using from Logistic

$\boxed{r>0}$ $N' = rN(1-N/k)$, Notice that $f(N) = rN(1-\frac{N}{k})$ and $f'(N) = r - \frac{2rN}{k}$.

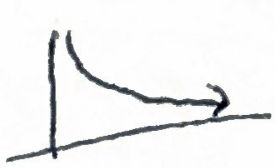



$$rN\left(1-\frac{N}{k}\right) = 0 \iff N^* = 0, \ N^* = K$$

$$f'(0) = r, \quad f'(K) = -r$$

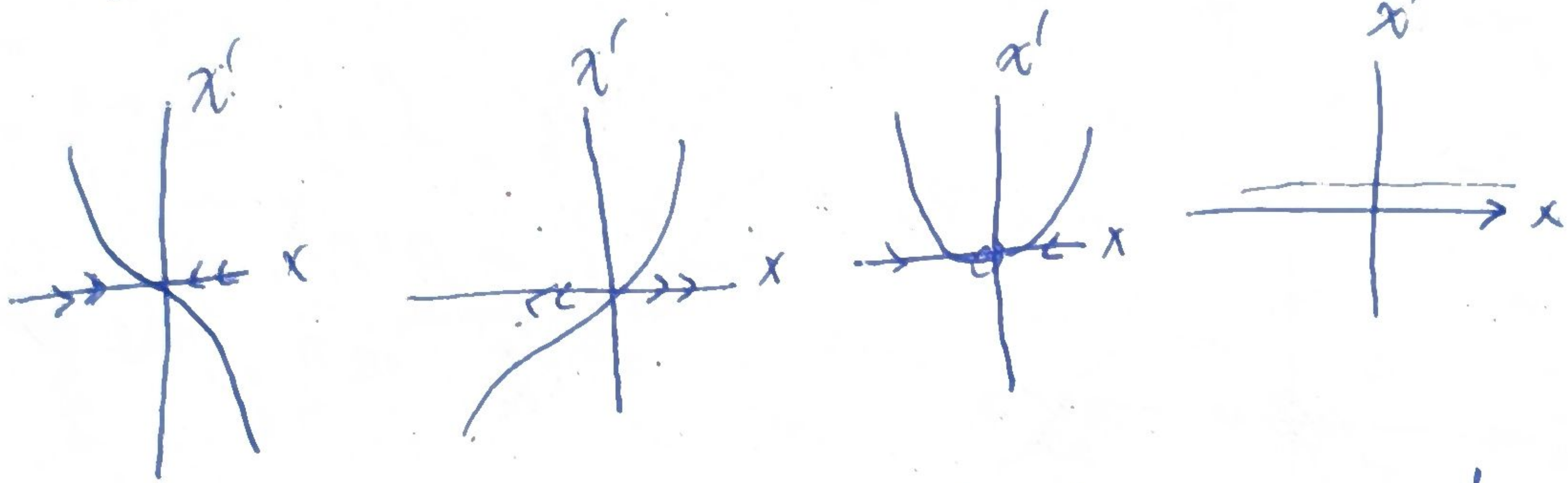
↑ UNSTABE ↑ STABLE.

$$1/|f'(N_i^*)| = \frac{1}{r}$$



Ex ¿Qué podemos decir sobre $f'(x^*)=0$?

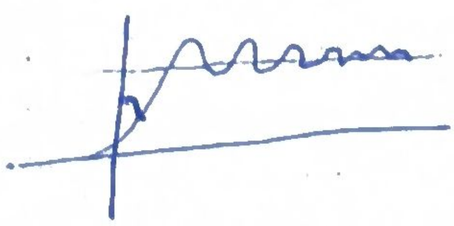
$x' = -x^3$, $x' = x^3$, $x' = x^2$, $x' = k$




$f'(x^*)=0$ does not mean that is a node.

IMPOSSIBILITY OF OSCILLATIONS

Notice that by now the solution or goes to an accumulation point or $\pm\infty$. The reason is that the solution are forced to increase or decrease monotonically (or remains constant). What does this means? The phase point never reverses the direction. This means that dumped/overshoot osc. could never occur.



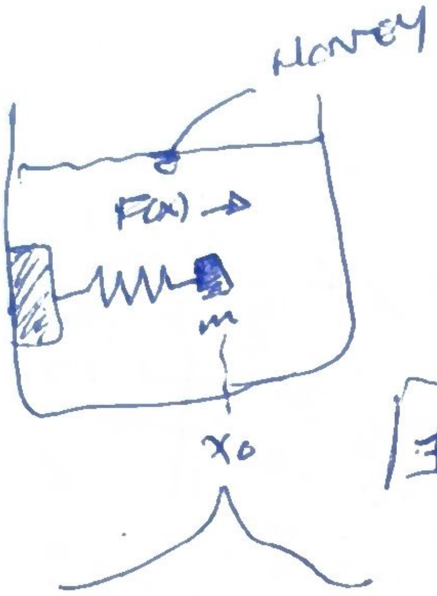
HENCE THERE ARE NO PERIODIC SOLUTIONS

LETS REMEMBER THAT WE ARE ON A STRAIGHT LINE (FLOW) THE FLOW COULD NO RETURN SUDDENLY.!

10

→ NOTICE THAT WE COULD FLOW ON CIR...

## EXAMPLE Mechanical Analogy




NOTICE THAT SPRING FORCE

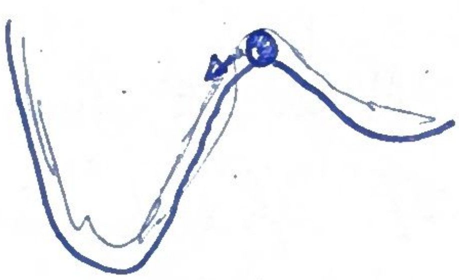
$$F(x) = m\ddot{x} + b\dot{x}$$

if $b\dot{x} \gg m\ddot{x}$

$$F(x) = b\dot{x} \to \frac{1}{b}F(x) = \dot{x} = f(x)$$

$$\boxed{f(x) = 0}$$

## EXAMPLE "Potentials"



There is a way to also visualize how the systems behave based on the physical idea of potential energy.

potential $V(x)$ $\qquad f(x) = -\frac{dV}{dx}$

$$V(x) := V(x(t))$$

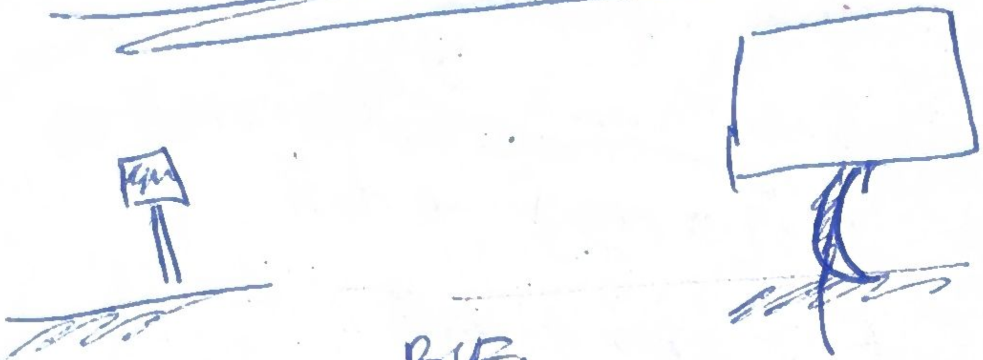
$$V' = \frac{dV}{dt} = \frac{dV}{dx}\frac{dx}{dt} = -\left(\frac{dV}{dx}\right)^2 \leq 0$$

↳ $V(x)$ DECREASES along trajectories, i.e. in particular always moves towards the lower potential. Notice that local minima of $V(x)$ are stable points, maximum are unstable points

$$\dot{x} = -x \quad , \quad \dot{x} = x - x^3$$

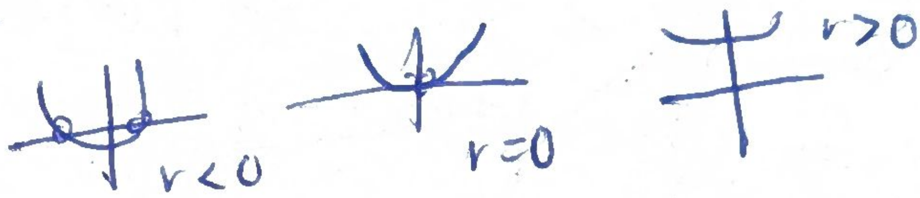
↳ $-\frac{dV}{dx} = -x \qquad -\frac{dV}{dx} = x - x^3$

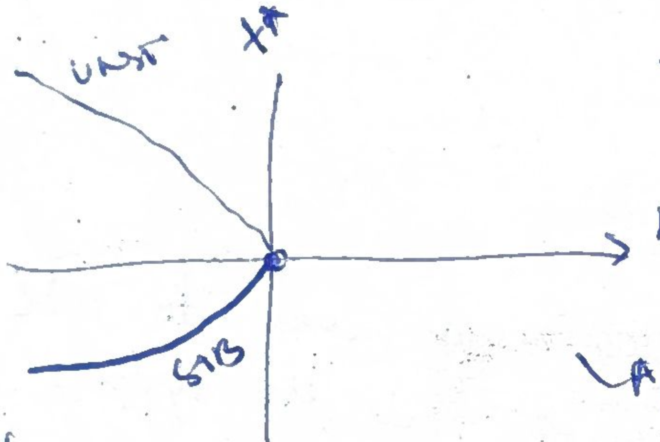
# BIFURCATIONS



BIF.

SADDLE - NODE $\qquad x' = r + x^2$




↳ Bifurcation Diagram

↳ $\pm\sqrt{r}$

blue sky bifurcation — Abraham & Shaw '88

"out of clear blue sky a pair of fixed points appear" $\quad x' = r - x^2$

$r < 0 \to$ NO EQ.P. , $r=0$ - 1, $r>0$ Two of them

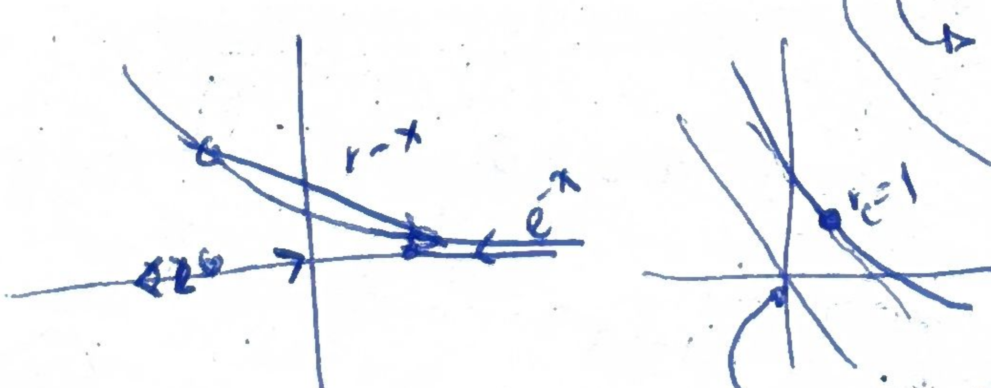
EXAMPLE $\qquad x' = r - x - e^{-x}$ $\qquad$ lets find derivt.

$r - x - e^{-x} = 0 \Rightarrow r - x = e^{-x} \to$ derivt.

↳ $\frac{d}{dx}(r - x) = \frac{d}{dx}(e^{-x})$

↳ $-e^{-x} = -1 \Rightarrow x^* = 0$

↳ $\boxed{r_c = 1}$



# Formas normales

In some way the family $x' = r \pm x^2$ are representative of all saddle-node bifurcation. We call them then "prototypical" and the main idea is that near enough to the eq point - in the range of the bifurcation, then it looks like $x' = r \pm x^2$

Let's consider our last example.

$x' = r - x - e^{-x}$; let's remember that the taylor serie around a point $a$ is

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \dots$$

[NOTE $a=0 \Rightarrow$ Maclaurin series].

$$e^{-x} = e^{a}\left(1 + (-x-a) + \frac{1}{2}(-x-a)^2 + \frac{1}{6}(x-a)^3 + \cdots\right]$$

$a=0$

$$e^{-x} = \left(1 - x + \frac{x^2}{2} + \cdots\right)$$

Then if we take our system and expand around $x=0$ (which ~~cod~~ is the Eq P. around $r=1$)

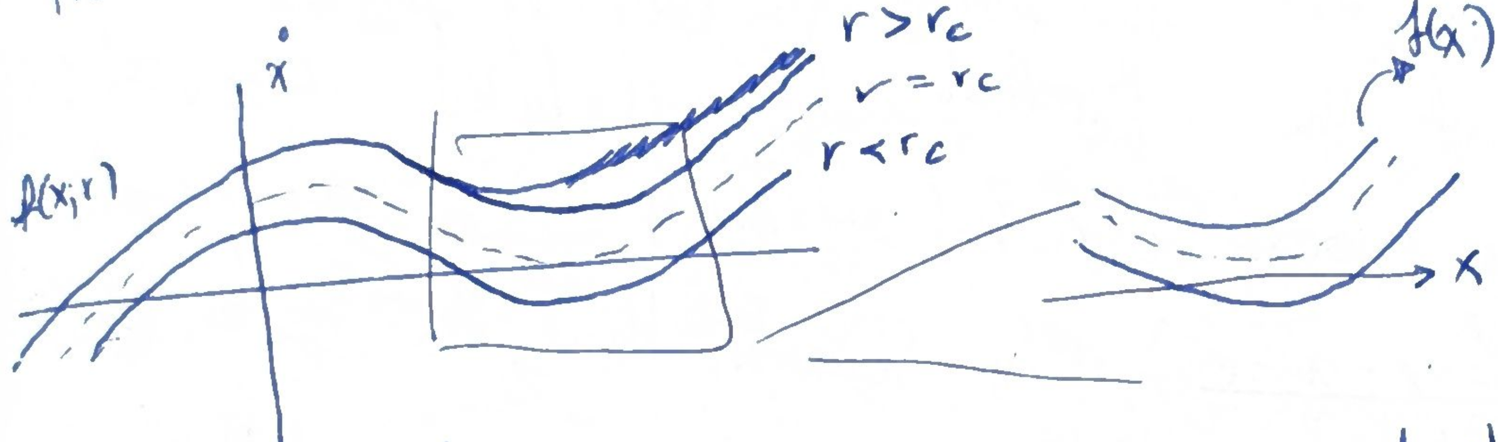
$$\text{then } x' = r - x - e^{-x} \approx r - x - \left[1 - x + \frac{x^2}{2} + \mathcal{O}(x^3)\right]$$

$$\approx (r-1) + x - x - \frac{x^2}{2}$$

$$\approx (r-1) - \frac{x^2}{2} + \mathcal{O}(x^3) \sim \underline{x' = r - x^2}$$

↳ SEEMS LIKE

It's easy to understand why this happens.
Graphically, we could expect these bifurcation always that $f(x)$ has/look locally as a bowl, this means that two roots are nearby ~~together~~




Algebraicly speaking, let $x' = f(x;r)$ be a cont dyn sys when $f(x^*) = 0$ and the critical parameter $r_c$.

1st let's remember that the Taylor function for a 2-dim function is

$$f(x_1,x_2) = f(a_1,a_2) + \left[(x_1-a_1)\partial_{x_1}f + (x_2-a_2)\partial_2 f\right] \Big|_{x_1=a_1,\, x_2=}$$
$$+ \frac{1}{2!}\left[(x_1-a_1)^2\partial_1^2 f + 2(x_1-a_1)(x_2-a_2)\partial_{12}^2 f + (x_2-a_2)^2\partial_2^2 f\right] + $$

Then our system.

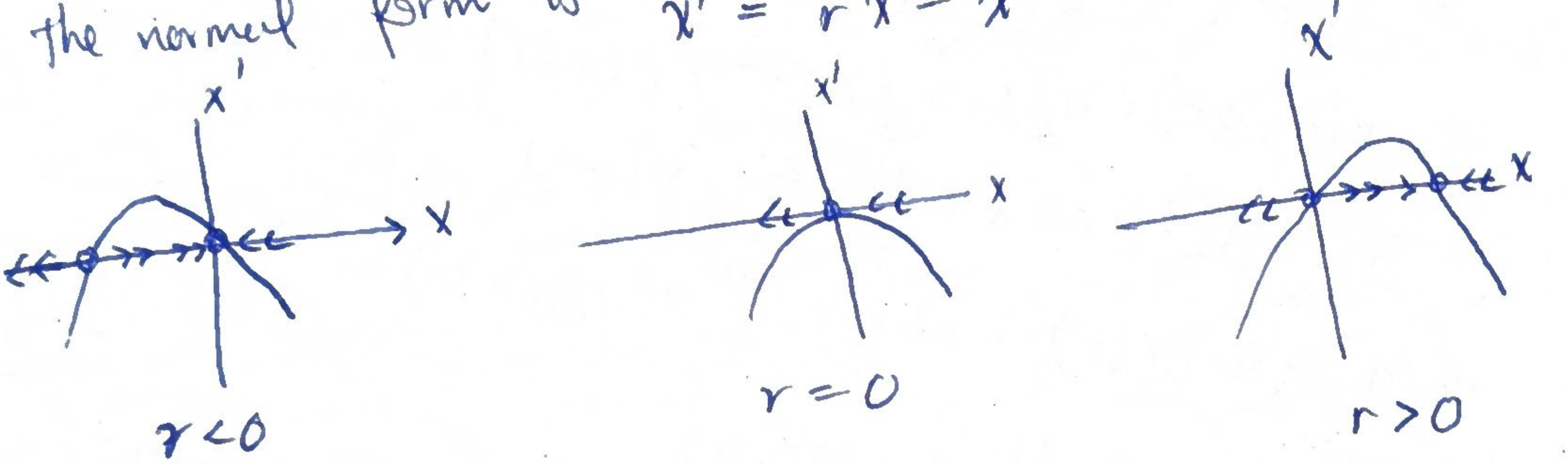
$$x' = f(x,r) \Big|_{x^*, r_c}$$
$$= \cancelto{0}{f(x^*, r_c)} + \left[(x-x^*)\partial_1 f^*(x^*,r_c) + (r-r_c)\partial_2 f(x^*,r_c)\right]$$
$$+ \frac{1}{2!}\left[(x-x^*)^2\partial_1^2 f(x^*,r_c) + \partial_2^2 f(x^*,r_c)(r-r_c)^2 + 2(x-x^*)(r-r_c)\partial_{x r}^2 f(x^*,r_c)\right.$$
$$+ O(r_c^3, a^3) \quad \Big| \text{Note } f(x^*,r_c) = 0,\ \partial_x f(x^*,r_c) = 0 \to \text{saddle-node}$$
$$= \partial_r f(x^*,r_c)(r-r_c) + \partial_x^2 f(x^*,r_c)(x^*-x^*)^2 + \dots$$
$$= a(r-r_c) + b(x-x^*)^2 + \dots +$$

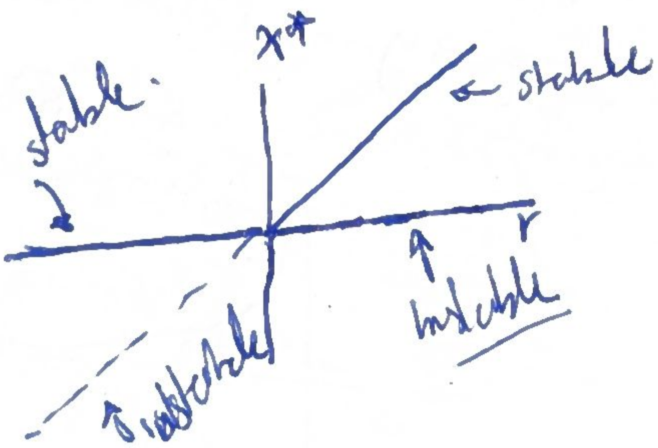
NORMAL FORM.

# Transcritical Bifurcations.

There are certain situations where a fixed point will never be destroyed. ~~For ex~~ E.g. the logistic equation & other simple models for the growth of a single species, there is a fixed point at $N=0$ but the stability could change as the ~~growth rate var~~ parameters varies (I called this "the context"), this kind of systems are ~~called~~ said to exhibit a transcritical bifurcation

the normal form is $x' = rx - x^2$




SOME PEOPLE SAY THIS is an exchange of stabilities

**Ex** Show that the system $x' = x(1-x^2) - a(1-e^{-bx})$ undergoes a transcritical bifurcation at $x=0$ when $a,b$ satisfy certain eq. (the eq ~~(a,b)~~ will define a bifurcation curve in the $(a,b)$ parameter space) Then find an approximated formula for the fixed point that bifurcates from $x=0$, for the bifurcation curve. assuming parameter $a,b$ near to the bifurcation curve.

**Sol** NOTICE THAT AROUND 0 THE EQ LOOK LIKE.

$$x' = x(1-x^2) - a\underbrace{(1-e^{-bx})}$$

$$= x(1-x^2) - a\left[\underbrace{1-(1}-bx+\tfrac{1}{2}b^2x^2+\theta(x^3))\right]$$

$$= x(1-x^2) - a\left[bx+\tfrac{1}{2}b^2x^2+\theta(x^3)\right]$$

$$= \underbrace{x(1-x^2)} - abx + \frac{ab^2}{2}x^2 + \theta(x^3)$$

$$= x - \underbrace{x^3+\theta(x^6)} - abx + \frac{ab^2}{2}x^2 + \theta(x^3)$$

$$= x - abx + \frac{ab^2}{2}x^2 + \theta(x^3)$$

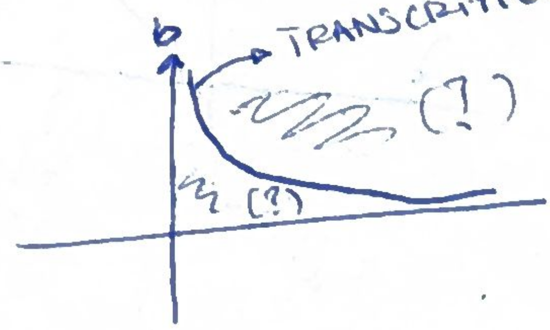
$$= \underline{(1-ab)x} + \frac{ab^2}{2}x^2 + \theta(x^3)$$

NORMAL FORM

$x' = Rx - x^2$

TRANSCRITICAL BIFURCATION

$$\{1-ab=0\} \to \{ab=1\}.$$




we know that.

$$x' \simeq (1-ab)x + \frac{ab^2}{2}x^2$$

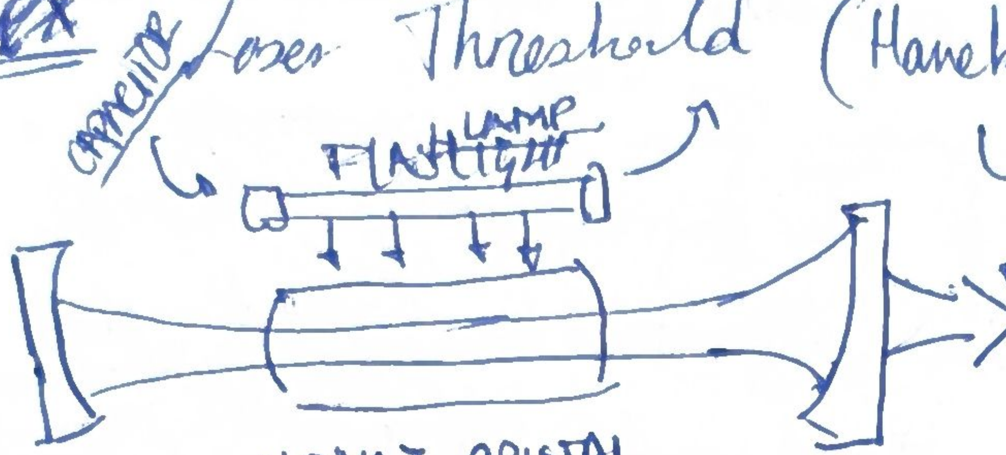
$$(1-ab)x^* + \frac{ab^2}{2}x^{*2} = 0$$

$$1-ab + \frac{ab^2}{2}x^* = 0$$

$$x^* \approx \frac{2(ab-1)}{ab^2}$$

→ NOTICE THAT ONLY VALID IF $x \ll 1$ & $ab \sim 1$.

~~EX~~ Laser Threshold (Haneken, 1983)




↳ LASER THEORY.

LASER OUTPUT

→ EXPLOTE

NO:YAG CRISTAL → Yttrium / Oxig / Alu

↳ ACTIVE MATERIAL.

The flashligh / pump excites the atoms on the material, once in excited the atoms ~~we~~ & returns to ~~basic~~ ground state and the emits Light. Then, we could think on atoms as antenas emiting energy? When the pumping is relatively weak the atoms travel at random but when ~~y~~ we start to increas the strength of pumping then. the ~~photons~~ start to oscilate on phase and we have a laser. Notice that, for having that effect of laser we need to traspass certain threshold.

This phenomenen is called synchronization / self-organizing.

Now a proper model would requires to know a bit of Eletrodynamics, Quantum Physics & bit of solid state physics (atomics). But we could extract the a model simplified from Haken, 1983 pp. 127-129) proposes a model where we measure

$n(t)$ " the number of photons in laser field ".

$$n' = \text{gain} - \text{loss} = \xi n N - kn.$$

gain: stimulated emission → light exciting atoms

NOTICE!: This is a random phenomena, ~~so is rer~~ this means that a rate proportional to $n(t)$ (the number of photons) & the number of $\underrightarrow{N(t)}$

$G > 0$ & is called "gain coefficient"

loss: the loss of photons is because the possible escape of photons either by laser or heat

$k > 0$ & is a rahara, $\tau = 1/k$ is the mean expentavey life ah laser.

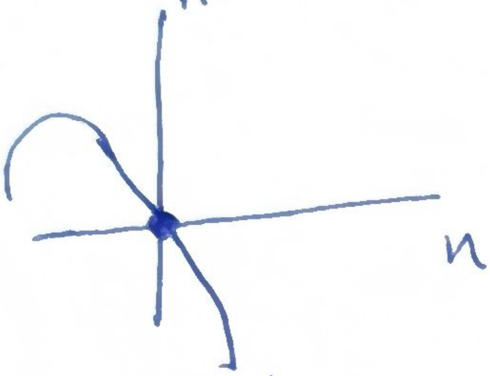
Now, after an atom gets excited then emits a photon and the returns to a grand state, we need then a eq that relates $N(t)$ with $n(t)$, $N(t)$ will decrease by emission of photons

Let's think only in one pulse of the pump, then we will have $N_0$ excited atoms and they will reduce by a factor by the laser process $N(t) = N_0 - \alpha n(t)$. , $\alpha > 0$

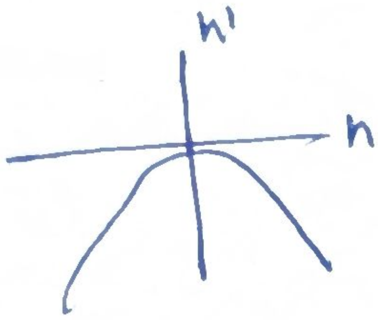
$\alpha$: the ratio at wich the atoms return to their grand state.

$$n' = Gn(N_0 - \alpha n) - kn = (GN_0 - k)n - \alpha G n^2$$

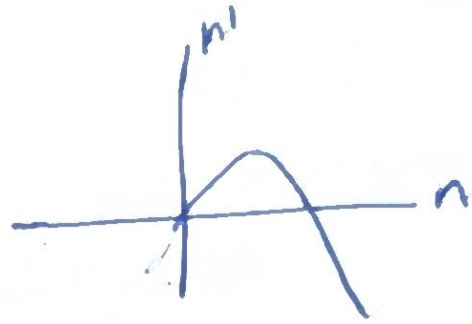
& finaly we are an familiar grand



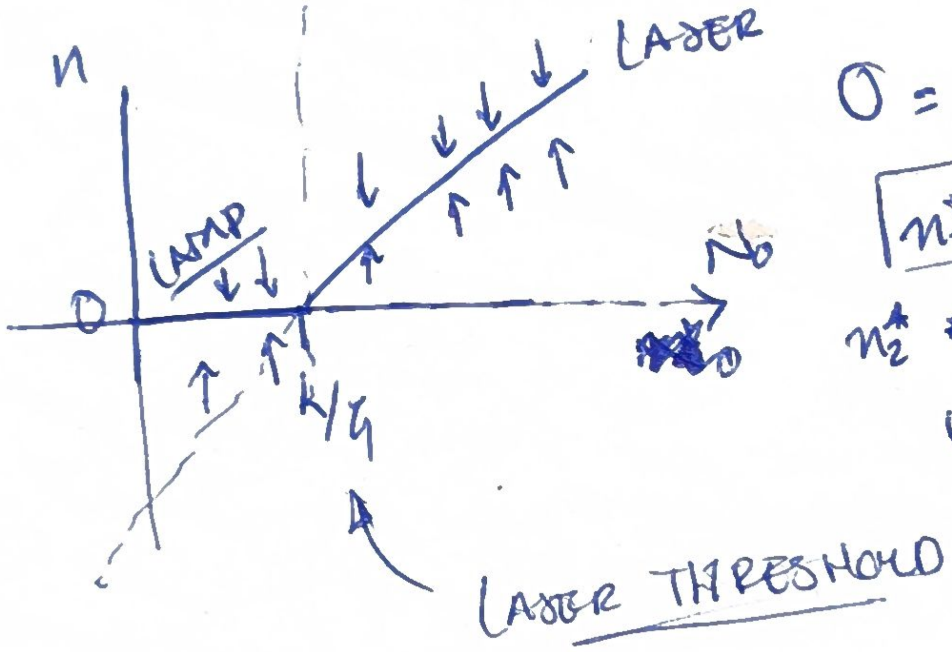

$N_0 < k/\eta$




$N_0 = \frac{k}{\eta}$




$N_0 > \frac{k}{\eta}$




$0 = n([\eta N_0 - k] - \alpha\eta n]$

$\boxed{n_1^* = 0}$

$\eta N_0 - k - \alpha\eta n = 0$

$n_2^* = \eta N_0 - k$

$\hookrightarrow \boxed{\frac{\eta N_0 - k}{\alpha\eta} = n_2^*}$